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Top-P

Top-P (Nucleus Sampling)

1. What *Top-P* Means

Top-P sampling—also called **nucleus sampling**—limits token choices to the **smallest set whose cumulative probability mass ≥ P**.

- **Smaller P** → **focused** (model sticks to the high-confidence “nucleus”).
- **Larger P** → **exploratory** (model reaches further into the probability tail).
- *Dynamic:* the number of candidate tokens changes every step, unlike the fixed **K** in Top-K.

2. How It Works Under the Hood

1. The model assigns a probability to every possible next token.
2. Tokens are sorted from highest to lowest probability.
3. Starting from the top, probabilities are summed until they exceed **P**.
4. All tokens inside this cumulative “nucleus” remain; the rest are discarded.
5. One token is sampled (often with temperature applied *within* the nucleus).
    - **P ↓** → Nucleus shrinks → output becomes safer, more deterministic.
    - **P ↑** → Nucleus grows → output gains diversity, but may drift.

3. Typical P Bands

Band	Output Character	When to Use
0.1 – 0.5	Focused, accurate, low variance	Math, data extraction, legal clauses
0.6 – 0.9	Balanced creativity vs. coherence	General chat, tutorials, summaries

0.9 – 0.99	Highly diverse, inventive, risk of off-topic	Brainstorming, fiction, poetry, slogan ideation

> Rule of thumb: Lower P for precision, higher P for imagination.

4. Choosing the Right P

4.1 Precision-Driven Tasks

- Formal explanations, API docs, medical facts
- Recommended **P = 0.1 – 0.3** (often pair with **temperature ≤ 0.3**)

4.2 Balanced Output

- Conversational agents, technical blog posts, UX copy
- Recommended **P = 0.6 – 0.85** with **temperature ≈ 0.5-0.7**

4.3 Creative or Divergent Ideation

- Short stories, jokes, marketing taglines
- Recommended **P = 0.9-0.98** and **temperature ≥ 0.8**
- Consider **Top-K off** or set very high (e.g., K = 100+) so Top-P is the primary filter.

5. Best-Practice Tips

1. **Combine with temperature:** Top-P shapes *scope*; temperature tweaks *randomness* inside that scope.
2. **Progressive tuning:** Start around **P = 0.9**; decrease for more rigor, increase for more flair.
3. **Safeguard factual content:** Keep **P ≤ 0.5** for sensitive or compliance-centric text.
4. **Batch sample & curate:** For ideation, generate several outputs at high P, then prune manually.
5. **Mind prompt length:** Short prompts + high P can derail faster; longer contextual prompts help anchor relevance.

6. Common Pitfalls

- **Overly broad nucleus:** Very high P (≈ 1.0) can resemble unconstrained sampling—expect hallucinations.
- **Misinterpreting P as percentage of tokens kept:** It's about cumulative probability mass, not token count.
- **Ignoring other controls:** Stop tokens, max_tokens, and system instructions still heavily influence output.

7. Quick Decision Flow

1. **Need airtight accuracy?** → *Yes* → Set **P ≤ 0.3**.
2. **Want a balanced, human-like style?** → *Yes* → Try **P ≈ 0.7**.
3. **Aiming for wild creativity?** → *Yes* → Use **P ≥ 0.9** and sample multiple completions.

Key takeaway: *Top-P is your adaptability dial.* Lower values keep the model firmly on-script; higher values let it wander into novel territory—ideal for ideation, but always validate against your task's tolerance for risk.